

Repair What Matters: Criticality-Bounded Revision Cones for Persistent Captured Worlds

SWAYAM SINGAL

Persistent captured-world systems rarely contain geometry alone. Over time they collect observations, reconstruction state, explicit surface and appearance data, GPU-published records, and temporal rendering history. A physical change can be small in space and still reach much farther through this computational state. Restricting the update too aggressively risks leaving stale dependencies behind; rebuilding everything avoids that problem, but also throws away locality even when most of the world has no bearing on the output we care about.

We present *Criticality-Bounded Revision Cones* (CBRC), implemented in MAVEB. The framework is fail-closed: regional recomputation is allowed only when the unrepaired exterior can be bounded for a stated quantity of interest. CBRC combines exact predecessor closure, conservative finite-change bounds where they are available, an explicit output-error contract, and a frozen work model; otherwise it returns FULL. The residual left outside the selected repair support is checked against an independently rendered full-after reference, so we test measured residual $\leq$ emitted bound $\leq$ declared tolerance rather than trusting the planner alone.

On the frozen public campaign of 60 revisions over four RGB/SfM scenes, CBRC chose 44 certified-local repairs and 16 full rebuilds. We observed no certificate violation, and the native and reference output-cone planners agreed in every case. Fifty-six source edits were already above the protected RGB tolerance before repair. The calibrated work model gives a median selected ratio of 0.36953 of FULL, or $2.706\times$ lower calibrated work. A separate five-case check on a pinned trained 3DGS representation produced four local repairs, one fallback, and no observed violation. The frozen end-to-end path still inspects almost the full Gaussian set, although exact sparse discovery reduces median inspection to 1% in isolation. These results support selective recomputation under the stated assumptions; they do not establish universal safety, global optimality, or wall-clock speedup.

CCS Concepts: • **Computing methodologies** → **Computer graphics; Rendering; Reconstruction.**

Additional Key Words and Phrases: persistent captured worlds, incremental reconstruction, 3D Gaussian Splatting, certified recomputation, change-aware rendering, dependency analysis

1 INTRODUCTION

Many 3D systems now live well beyond the capture session that created them. A reconstructed room may be revisited after furniture has moved; an AR map continues to exist while objects appear and disappear; a digital twin is updated after a small physical intervention. The stored state in these systems is not one uniform representation. Observations feed geometry, geometry determines surface or Gaussian support, materials and textures follow ownership and visibility, GPU records mirror persistent entities, and temporal filters reuse earlier screen-space state. For that reason, an edit that looks

local in Euclidean space may not be local at all once it propagates through the system.

The safest response is still a global rebuild. Classical volumetric fusion and dense reconstruction give a natural full-reference notion of recomputation [5, 6, 17]. That choice is straightforward to reason about, but it becomes wasteful when only a small dependency closure is actually affected. Explicit scene representations make this mismatch particularly visible: large parts of a Gaussian map, or of a neural/hybrid reconstruction, may have no reason to change after a local revision [11, 12, 15]. Continual Gaussian systems already exploit this fact by focusing optimization on changed regions or by managing stale observations over time [2, 26, 27]. Our concern is slightly different. Those systems mainly decide *where to update*; here we ask how much error may safely remain outside the region that was chosen, across several kinds of derived state at once.

The question is deliberately narrow: *when is it defensible to repair less than the whole captured world while keeping an explicit output-error contract?* Reachability alone is not enough. Ownership, record identity, publication, and unstable temporal invalidation are structural relations that should not be approximated. Other relations, such as edited Gaussian support or stable temporal blending, have finite influence that can be bounded. The revision policy must distinguish the two and fall back to a full rebuild when the remainder outside a proposed repair region cannot be bounded tightly enough.

Our answer is *Criticality-Bounded Revision Cones* (CBRC). A revision is represented on a typed dependency graph with three edge meanings. HARD edges carry exact structural requirements; ANALYTIC edges carry conservative finite-change bounds; and EMPIRICAL edges may help decide what to try first, but they never become part of the safety certificate. For a proposed cone C , we certify the unrepaired exterior O against the declared quantities of interest. The worst normalized certificate-to-tolerance ratio is the *revision criticality*, with one as the decision boundary. We accept a local result only when exact predecessors are closed, every protected output is at or below that boundary, and the calibrated regional work is below the independently measured full baseline. If the coupling is unsupported, the residual is supercritical, the analytic assumptions are unstable, or locality simply costs too much, the planner returns FULL.

MAVEB implements this design as a C++23 persistent-world stack, alongside a dense Python reference certifier and an independent full-after Gaussian oracle. Figure 1 summarizes the decision path, while Figure 2 shows the support-selection idea visually. The implementation records

Author’s address: Swayam Singal, swayam.singal@gmail.com.

immutable before/after revision sidecars, deterministic certificate JSON, per-domain work, and oracle replay. We do not treat the planner’s own prediction as evidence that a repair is safe. Instead, the unrepaired exterior implied by the chosen support is compared with an independently rendered FULL-after reference and checked under the contract

$$E_q^{\text{full-ref}} \leq B_q(C) \leq \varepsilon_q, \quad (1)$$

where $E_q^{\text{full-ref}}$ is the measured residual attributable to the unrepaired exterior under the selected support, evaluated against an independent full-after rendering; $B_q(C)$ is the emitted conservative bound, and ε_q is the declared tolerance for quantity of interest q .

We froze the protocol and claim boundaries before interpreting the final campaign. Among 60 public RGB/SfM revisions, CBRC makes 44 certified-local selections and 16 full rebuilds, with no observed certificate violation and complete native/Python output-cone planner agreement. Fifty-six source edits were above RGB tolerance before repair. The frozen calibration gives a median work ratio of 0.36953 of FULL, or $2.706\times$ lower calibrated work; this is not a paired wall-clock speedup. On the pinned trained 3DGS representation, four of five revisions stay local and one falls back, again without an observed violation. Candidate discovery, however, is still almost global in the frozen path.

The contributions are:

- a fail-closed formulation for revising heterogeneous persistent captured worlds, combining exact dependency closure with conservative finite-change propagation;
- output-specific certification for local repair, including implementation-backed Gaussian display-space and stable temporal bounds, while unsupported dependencies stay exact;
- a native work-aware planner with automatic FULL fallback, deterministic evidence emission, a dense reference implementation, and an independent full-after falsification path; and
- frozen evidence on public and trained representations that reports the local/full crossover, calibrated work, sparse-discovery behavior, neutral or negative ablations, and the limitations we actually observed, without relabeling work reduction as speedup.

2 RELATED WORK

Persistent reconstruction and map maintenance. Volumetric integration gave reconstruction systems a durable way to accumulate range observations [5], and surfels offered an explicit point-oriented alternative [19]. KinectFusion later showed real-time dense fusion and tracking [17]; ElasticFusion maintained and corrected dense surfel maps without a conventional pose graph [22]; and BundleFusion reintegrated surfaces online while maintaining global consistency [6]. All of these systems make map state persistent enough that revision becomes meaningful, although their update rules are

closely tied to the reconstruction or consistency mechanism itself. CBRC looks at the revision one level above that: which derived state really has to be recomputed, what may remain outside it, and at what point should the incremental attempt be abandoned.

Neural and Gaussian mapping. NeRF established continuous neural radiance fields for novel-view synthesis [16], while iMAP and NICE-SLAM brought learned implicit representations into online mapping and SLAM [21, 28]. 3D Gaussian Splatting (3DGS) moved back toward an explicit anisotropic representation while retaining real-time rendering [12]; SplatTAM and Gaussian Splatting SLAM then adapted Gaussian state to online RGB-D and monocular mapping [11, 15]. Dynamic 3D Gaussians and 4D-GS extend the representation over time [14, 24]. These representations make revision especially tangible because individual persistent primitives and their support are exposed. Still, the CBRC formulation itself does not depend on Gaussian state.

Continual and evolving Gaussian scenes. CL-Splats separates changed and static scene components so that Gaussian optimization can stay focused while older scene states are preserved [2]. GaussianUpdate uses change-specific update strategies and visibility-aware replay for continual radiance fields [27]. GaME addresses long-term evolving RGB-D scenes by adapting the Gaussian map while managing stale observations [26], and EliGSiR studies continual Gaussian mapping under a bounded compute budget through adaptive view scheduling, supervision fidelity, and geometry growth [7]. We therefore do not claim the broad idea that changing Gaussian scenes should be updated locally. The distinction in CBRC is the contract: cross-representation exact dependencies and conservative output bounds are checked together, locality is admitted only under Eq. 1, and FULL remains an explicit outcome when either certification or the work comparison fails.

Editing, change detection, and identity. GaussianEditor and SC-GS demonstrate precise, controllable local editing of Gaussian representations [4, 10]. Primitive-space change detection further suggests that Gaussian attributes themselves can reveal structural and appearance changes [8], while Consistent Instance Field studies persistent probabilistic identity in dynamic Gaussian scenes [25]. These are close enough to our setting that the boundary needs to be explicit: we are not claiming local editing, primitive-level change detection, or persistent identity as new. CBRC starts after a revision has been specified or detected and asks what can safely remain unrepaired for a particular output.

Incremental rendering, change propagation, and goal-oriented error control. Dependency-aware recomputation is not new to graphics. Bala et al. connect edited world regions to bounded-error radiance interpolants with ray segment trees [3], and Wörister et al. derive a dependency graph from scene-graph changes to rendering caches so those caches can be updated incrementally [23]. Beyond graphics,

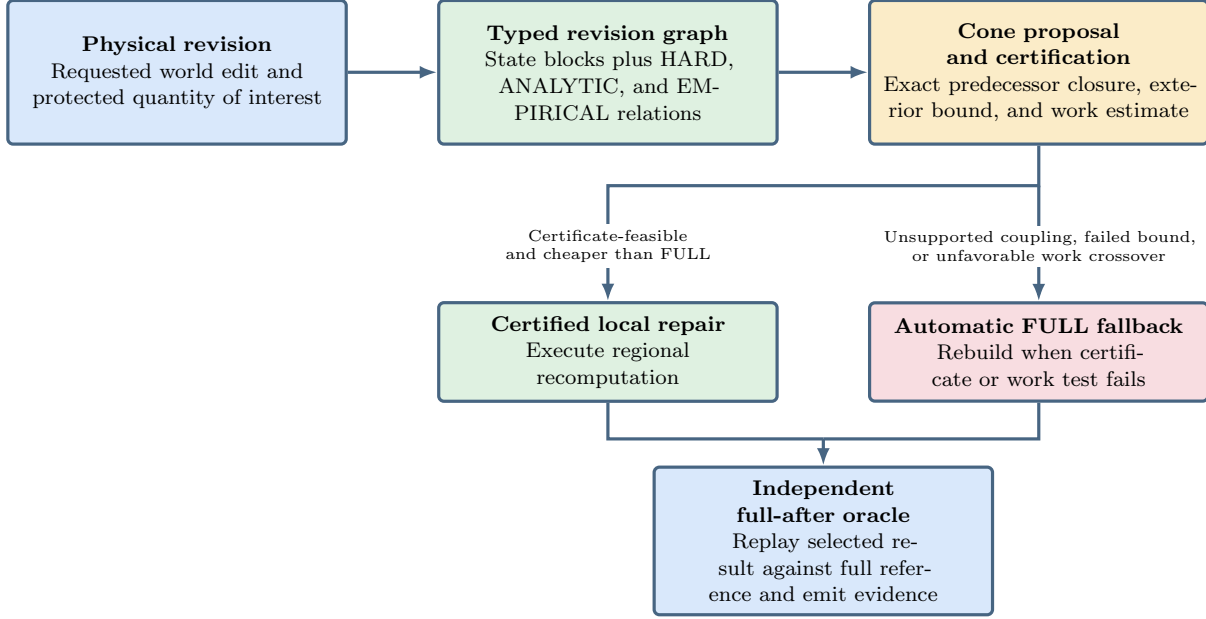


Fig. 1. CBRC decision and falsification path. A physical revision enters a typed dependency graph. Exact HARD predecessors are closed, conservative ANALYTIC influence is propagated to the unrepaired exterior, and quantity-of-interest bounds are checked against tolerance. A local repair is executed only when it is certified and cheaper than the full baseline; otherwise the system falls back to FULL. The residual outside the selected repair support is then falsified against an independently rendered full-after reference.

self-adjusting computation formalizes correct change propagation under mutation and reuse [1]; goal-oriented error estimation directs numerical work toward selected quantities of interest rather than a global state norm [18]. Several ingredients of CBRC clearly live in these traditions, so we treat them as novelty boundaries rather than background to be glossed over.

The mathematics used by CBRC is mostly classical. Graph closure, non-negative influence propagation, Neumann-series resolvents, norm bounds, and greedy search are not contributions of this paper. What we claim instead is a systems contract for captured worlds that brings these pieces together around one revision decision: exact dependencies across derived state, conservative finite-change bounds, rendered-output quantities of interest, a heterogeneous work crossover, fail-closed FULL fallback, and independent full-reference falsification. The mathematics is the machinery that makes that contract executable, not a standalone novelty claim.

3 PROBLEM FORMULATION

3.1 Persistent captured-world state

Let the current system state be partitioned into blocks

$$V = \{1, \dots, n\}. \quad (2)$$

A block can represent an observation region, TSDF block, mesh patch, texture page, material state, Gaussian subset, GPU publication range, temporal-history region, or another

Table 1. Revision-edge semantics in CBRC.

Type	Meaning	Certificate role
HARD	Exact ownership, identity, topology, publication, or invalidation dependency	Must be closed exactly
ANALYTIC	Conservative implementation-backed finite-change upper bound	May certify exterior influence
EMPIRICAL	Measured or learned influence used for ordering or scheduling	Never treated as proof

unit of derived state. It is a unit of computational ownership, not necessarily a regular spatial cell.

A physical revision gives a direct change envelope $b \in \mathbb{R}_{\geq 0}^n$. From there, the planner chooses a repair cone $C \subseteq V$ and leaves the exterior

$$O = V \setminus C \quad (3)$$

untouched. Small cones are useful, but minimizing $|C|$ by itself is not the objective. The cone has to remain feasible under the output certificate, and its measured or calibrated work must still beat the corresponding full rebuild.

3.2 Typed dependencies

Each dependency has one of three semantics (Table 1).

How Local Gaussian Revisions Propagate

From intended physical edit to certified local repair or FULL fallback.

MAVEB / CBRC
Certified Local Revision of
Persistent Gaussian Worlds

Edit locally.
Prove the residual.
Fall back to FULL
when necessary.

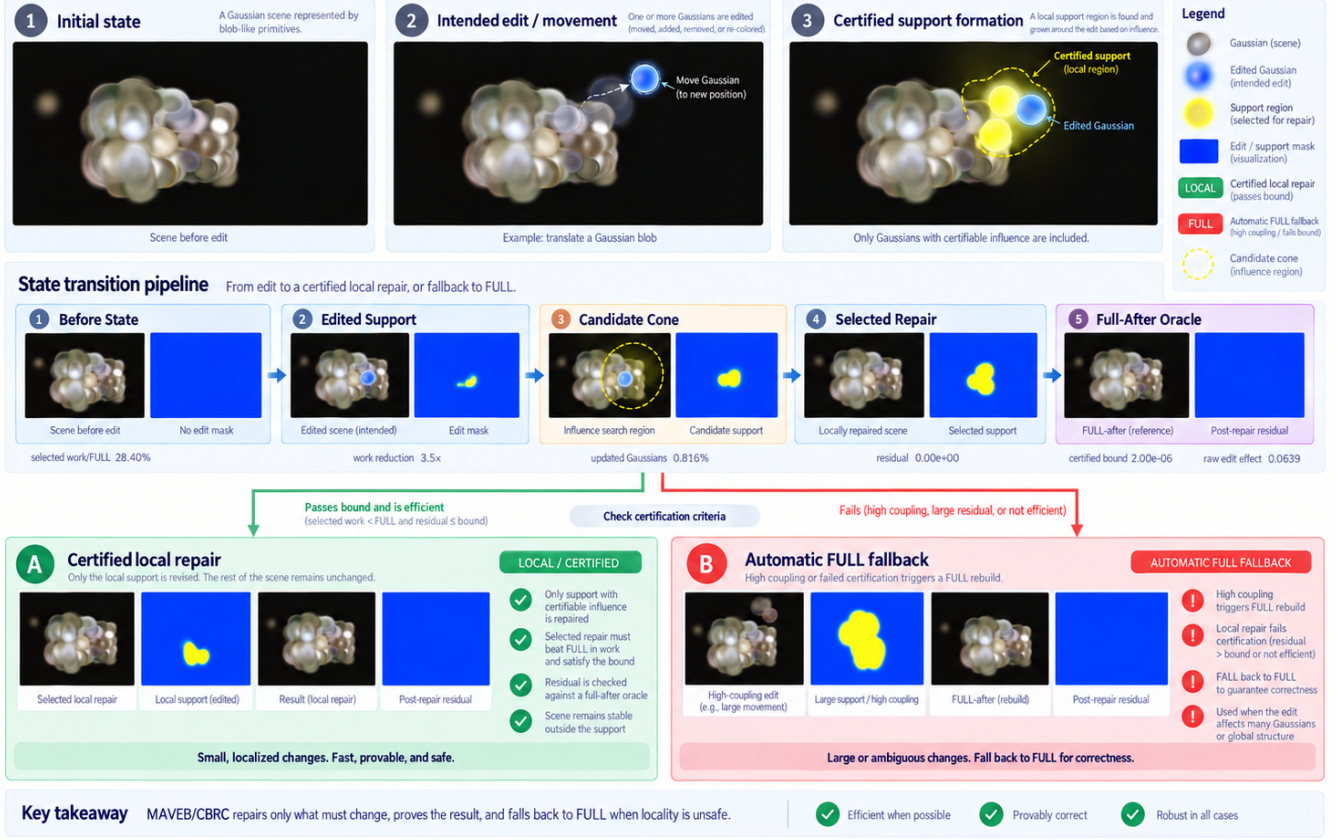


Fig. 2. How a physical edit becomes a certified revision. An intended Gaussian edit induces a candidate influence region from which CBRC selects the support that must be repaired. Exact structural dependencies remain closed and the unrepaired exterior is accepted only when its quantity-of-interest bound satisfies the declared tolerance and regional work remains below the frozen FULL baseline. The system therefore either revises only the certified support (A) or fails closed to a FULL rebuild (B). Blue marks the intended edit, yellow the selected repair support. The supplied image is reproduced unchanged. Its 28.40% work ratio, 3.5 $\times$ work reduction, and other embedded micro-metrics are illustrative, not aggregate campaign results. The frozen public headline is 0.36953 of FULL (2.706 $\times$ lower calibrated work; Section 7). Statements of correctness in the image apply only under the bound assumptions and declared revision domain; neither these work factors nor the image establish wall-clock speedup or universal safety.

We keep this separation conservative on purpose. Dependencies from observations to TSDF state, TSDF to mesh ownership, meshes to texture-page identity, and Gaussian records to GPU publication remain exact in v1. Unstable temporal validation is treated the same way. By contrast, Gaussian-to-image influence and stable temporal blending have implemented ANALYTIC bounds. An EMPIRICAL-only dependency without a conservative analytic bound is promoted to exact predecessor closure for certification; its measured gain may still be used for scheduling, but never as a soft safety bound. If we do not have a conservative argument for a relation, we do not soften it simply because doing so would produce a smaller cone.

3.3 Output and work contracts

For each quantity of interest q , let $\varepsilon_q \geq 0$ denote its declared tolerance. A local repair is admissible only when $B_q(C) \leq \varepsilon_q$ for every protected output. That is the planner-side requirement. The experiment then asks the harder question by checking the independent full-reference contract in Eq. 1.

Locality also has to be worth doing. The planner therefore requires regional work to remain below FULL. Because the native counters describe different kinds of work, adding them directly would be meaningless. If $n_d(C)$ is native work in domain d and κ_d is the frozen calibrated cost of one native

unit, the public campaign uses

$$W(C) = \sum_d \kappa_d n_d(C), \quad (4)$$

with W_{full} measured independently. The calibration is frozen before the final campaign. We keep this distinction explicit throughout the paper: Eq. 4 models heterogeneous work; it is not an end-to-end timing measurement.

4 CRITICALITY-BOUNDED REVISION CONES

4.1 Exact predecessor closure

If a repaired state block v requires an exact predecessor u , then the predecessor must also be inside the cone:

$$v \in C, u \in \text{Pred}_{\text{HARD}}(v) \implies u \in C. \quad (5)$$

This condition is checked before any approximate reasoning. If Eq. 5 is violated, the candidate is invalid regardless of how small a local sensitivity estimate happens to be.

4.2 Conservative exterior propagation

For ANALYTIC dependencies we use a componentwise non-negative influence matrix $K \in \mathbb{R}_{\geq 0}^{n \times n}$. Each entry K_{vu} upper-bounds the normalized finite change that block u can transmit to block v . Let z_C collect known change bounds for the repaired state. The unrepaired exterior then satisfies

$$\delta_O \leq b_O + K_{OC} z_C + K_{OO} \delta_O. \quad (6)$$

When the exterior response is certifiable and $\rho(K_{OO}) < 1$, the classical resolvent gives

$$G_O = (I - K_{OO})^{-1} = I + K_{OO} + K_{OO}^2 + \dots, \quad (7)$$

and therefore

$$\hat{\delta}_O = G_O (b_O + K_{OC} z_C), \quad \delta_O \leq \hat{\delta}_O. \quad (8)$$

The native v1 planner does not evaluate a general cyclic resolvent. It uses sparse finite propagation on supported DAG exteriors. If an analytic cycle falls outside that supported case, the planner fails closed: either the cone expands until the unsupported exterior disappears, or the result becomes FULL. Equation 8 should therefore be read as the reference formulation, not as a claim that arbitrary cyclic systems are handled in production.

4.3 Quantity-of-interest certificate

Let R_q map state perturbation to the protected output quantity. The exterior contribution is bounded by

$$B_q(C) = \left\| |R_{q,O}| \hat{\delta}_O \right\|_{\infty}. \quad (9)$$

A candidate is certificate-feasible only when $B_q(C) \leq \varepsilon_q$ for every declared output. As a result, the same geometric change can lead to different revision decisions when the protected outputs differ. This is intentional: the certificate is tied to what the user actually asks us to preserve, rather than to an internal state norm that may or may not correlate with visible error.

4.4 Revision criticality

We use *revision criticality* for the worst normalized certified residual of a candidate cone

$$\begin{aligned} \chi(C) &= \max_q r_q(C), \\ r_q(C) &= \begin{cases} B_q(C)/\varepsilon_q, & \varepsilon_q > 0, \\ 0, & \varepsilon_q = B_q(C) = 0, \\ +\infty, & \varepsilon_q = 0 < B_q(C). \end{cases} \end{aligned} \quad (10)$$

The decision boundary is $\chi(C) = 1$. Values at or below one are certificate-feasible; a value above one means that at least one protected output is supercritical and the cone must grow or fall back to FULL. We use the phrase *criticality-bounded* in this narrow sense of a normalized output-error boundary. It is not meant to suggest a new model of physical critical phenomena.

4.5 Gaussian display-space bound

For a projected Gaussian i at pixel p , MAVEB starts from the projected opacity model

$$\alpha_i(p) = o_i \exp\left(-\frac{1}{2} q_i(p)\right), \quad (11)$$

where o_i is peak opacity and $q_i(p)$ is the squared Mahalanobis distance in projected Gaussian space. In the bound that follows, α_i means the effective value after the renderer's clamp and cutoff rules, not simply the raw value from Eq. 11. For an edited set E , we define the aggregate opacity mass

$$A_p(E) = 1 - \prod_{i \in E} (1 - \alpha_i(p)). \quad (12)$$

Proposition 1 (Gaussian finite-edit display bound). Let U denote the unchanged Gaussian state and E_0, E_1 the complete before/after edited sets at a protected pixel p . Under the assumptions below,

$$\begin{aligned} \Delta_p &= \|R_p(U \cup E_0) - R_p(U \cup E_1)\|_{\infty}, \\ \Delta_p &\leq C_{\text{color}} \min(1, A_p(E_0) + A_p(E_1)). \end{aligned} \quad (13)$$

Assumptions. The proposition rests on four conditions. First, compositing is front-to-back, with effective post-clamp/post-cutoff alpha in $[0, 1]$. Second, each protected color channel, including the background and the renderer's bounded spherical-harmonic evaluation, lies in an interval of width C_{color} . Third, E_0 and E_1 include every Gaussian whose projected opacity, color, insertion/deletion state, or depth relation to unchanged support can change because of the edit. Finally, renderer behavior that is not captured by these bounded quantities is treated as unsupported and is promoted to an exact dependency or a FULL fallback.

Proof sketch. For a single edited set, front-to-back compositing can alter a channel by no more than the channel range times the opacity mass through which edited support replaces, attenuates, or reveals unchanged radiance. This gives $C_{\text{color}} A_p(E)$. Insertion and deletion are the special cases in which one edited set is empty. For a general before/after

edit, we pass through the common unchanged state U and apply the triangle inequality, obtaining $C_{\text{color}}(A_p(E_0) + A_p(E_1))$. The result is then capped by the largest possible channel-range change, C_{color} . The image-level certificate takes the maximum of Eq. 13 over the declared protected pixels. The supplement gives the algebra in more detail.

4.6 Stable temporal history

For stable temporal validation, let E_c denote current-frame error, E_h retained-history error, E_n neighborhood or clamping error, and $w \in [0, 1]$ the history weight. With $E_r = \max(E_h, E_n)$,

$$E_{\text{resolved}} \leq (1 - w)E_c + wE_r. \quad (14)$$

If validation, reprojection, or disocclusion can itself change, MAVEB does not apply Eq. 14; it promotes the relation to HARD invalidation. The temporal case is a useful example of the design rule we follow throughout: a soft bound is used only for as long as its own preconditions can be certified.

4.7 Greedy feasible-cone search and crossover

The v1 search still needs a numerical ordering when a declared tolerance is zero. For stable exteriors it uses $\tilde{\chi}(C) = \max_q B_q(C) / \max(\epsilon_q, 10^{-15})$ only as a search score. The floor does not relax acceptance, which still uses $B_q(C) \leq \epsilon_q$. For an expansion $C \rightarrow C'$ with positive added work, the utility is

$$\text{utility}(C \rightarrow C') = \frac{\tilde{\chi}(C) - \tilde{\chi}(C')}{W(C') - W(C)}. \quad (15)$$

Each candidate is closed over its exact predecessors before it is certified again. The search continues until it finds a cone that is both certificate-feasible and cheaper than FULL; if no such cone is reached, the planner returns the full fallback. This procedure gives us a certified feasible cone. We do not claim that it is the globally cheapest safe cone.

5 SYSTEM IMPLEMENTATION

MAVEB keeps the scientific reference path separate from the production-oriented execution path. The dense Python certifier follows the mathematical contract directly and serves as a falsification and regression reference. The native C++23 implementation uses a sparse typed revision graph, sparse analytic propagation over supported DAG exteriors, exact predecessor checks, greedy feasible-cone search, and deterministic certificate serialization. For the frozen campaign, the bound production output-cone graph is evaluated by both the native planner and the Python reference planner; their final decisions are compared rather than assumed equivalent.

Revision transactions. The production path records persistent Gaussian entity translations, immutable before/after sidecars, exact camera binding, a production image certificate, and the planner’s temporal repair or retention decision. In headless operation, the system loads a persistent world, applies indexed Gaussian transactions, writes the new durable revision and sidecars, evaluates the Gaussian certificate, runs

the output-cone planner, and emits canonical evidence JSON. If the consistency checks do not pass, the revision fails closed instead of publishing a partly updated world.

Cross-derived-state accounting. A **LocalityLedger** tracks observation, TSDF, mesh, texture-page, material, Gaussian, GPU-publication, and temporal work; ledger/mesher disagreement is an error. Counts retain their native units until a versioned calibration is applied. The frozen public campaign calibrates only four measured domains: Gaussian inspection, Gaussian update, GPU publication bytes, and temporal invalidation pixels.

Independent oracle. The Gaussian oracle is independent of the planner. It renders before and full-after states, then forms a support replay from full-after values on the selected support and before-state values outside it. The per-pixel exterior residual is checked against Eq. 1; certificate disagreement or a residual above the emitted bound is a correctness failure. Evidence bundles record hashes, workflow IDs, and native/reference output-cone parity.

Representation scope. The public campaign uses four real scenes reconstructed with COLMAP/SfM [20], from which we seed isotropic Gaussians at the reconstructed points. We do not describe these fields as trained photorealistic 3DGS, and the public coordinates are normalized rather than metric-scale. A second validation path uses a pinned external trained GraphDECO-style PLY and preserves its spherical-harmonic coefficients, opacity, anisotropic scale, and rotation. Ownership in this path is deterministic and spatial; it should not be read as semantic segmentation.

6 EVALUATION

6.1 Questions and protocol

The evaluation asks seven questions: certificate validity (RQ1), local-versus-full crossover (RQ2), calibrated work (RQ3), measured-domain locality (RQ4), baseline behavior (RQ5), transfer to trained 3DGS (RQ6), and exact sparse discovery (RQ7).

The final public v2.1 campaign contains 60 frozen revisions over Tanks & Temples Train/Truck [13] and Deep Blending Dr Johnson/Playroom [9] inputs from the pinned GraphDECO bundle [12]. The repository manifest fixes the source archive size and SHA-256, and the work calibration is frozen before this campaign is run. We also keep an earlier execution in which a requested translation fell below the production effective-edit threshold. Rather than retuning or deleting that case, we preserve it as protocol history; v2.1 enforces the production precondition and records both requested and applied edit deltas.

For each selected repair, the campaign stores the source edit, chosen cone, decision, certificate, support-replay full-reference residual, native and calibrated work, and provenance. Fallbacks stay in the dataset and in the summary statistics; they are part of the method’s behavior, not failed trials to be filtered out.

Runtime scope. The current campaign does not include paired end-to-end LOCAL-versus-FULL wall-clock trials under identical execution conditions. We therefore make no speedup claim. The public headline is a frozen four-domain work ratio calibrated from isolated millisecond measurements, whereas the baseline suite reports a different native output-cone work-ratio metric. We label the two separately because collapsing them into one number would overstate what was actually measured.

6.2 Baselines and ablations

The method-level suite includes FULL, exact-only repair, fixed-radius policies, a changed-fraction policy, a held-out empirical scheduler, and CBRC. Empirical magnitudes may change the order in which candidates are explored, but they are never used as soft certificate evidence; an empirical-only dependency without an analytic bound fails closed through exact predecessor closure. Structural and analytic ablations remove or replace individual components. Separately, a deterministic mechanism-isolation suite creates stress cases for exact closure, analytic Gaussian influence, temporal influence, quantity-of-interest specialization, hard/soft separation, graph propagation, and fallback behavior.

We do not retune the real campaign just to make every ablation look different. Several structural ablations are neutral on the frozen matrix, and we report them that way. The synthetic mechanism suite serves a different purpose: it shows that a mechanism can be made consequential in a controlled stress case, not that the same mechanism determines effect size for every real revision.

6.3 Metrics

Our primary correctness metric is the number of violations of Eq. 1. We also report local/fallback counts, native/reference output-cone planner agreement, source-edit effect, the median calibrated work ratio, and native-domain ratios. Effectivity, $\eta = B_q/E_q^{\text{full-ref}}$, is only meaningful when the independent residual is non-zero. When the measured residual is numerically zero, we leave effectivity undefined rather than manufacture a finite value with an arbitrary denominator floor.

7 RESULTS

7.1 Public RGB/SfM campaign

Across the 60 frozen public revisions, CBRC chose a certified local repair 44 times and returned FULL 16 times. Figure 3 shows examples of both outcomes, and Table 3 collects the aggregate numbers. We observed no certificate violation, and the native and reference output-cone planners agreed on every decision. At the reported L_∞ precision, the largest support-replay residual outside the selected repair support was 0; the largest emitted bound among selected cases was 2×10^{-6} .

A zero support-replay residual would be less informative if the edits were visually negligible to begin with, so we checked

that separately. Before repair, 56 of the 60 source edits were already above the protected RGB tolerance. This rules out the simplest explanation that almost all edits were below the quantity-of-interest threshold anyway.

The frozen isolated millisecond cost model reports a median selected four-domain work ratio

$$\text{median } W(C)/W_{\text{full}} = 0.3695301, \quad (16)$$

which corresponds to $2.706\times$ lower calibrated work. We avoid calling this a $2.706\times$ speedup. The public scalar is assembled from frozen per-domain microbenchmarks, whereas campaign phase wall times are recorded separately. A speedup claim would require a paired local-versus-full end-to-end timing experiment under the same conditions.

7.2 Method comparison

Table 2 compares the method-level baselines on the same 60 frozen cases. Every selected cone is re-certified using the conservative certificate, so *certificate pass* should not be confused with an independent measured-error metric. The exact-only, radius, and changed-fraction policies often select very little work, but most of those selections do not pass the certificate. The empirical-order baseline uses empirical magnitudes only to decide expansion order; the final candidate is still certified with analytic bounds plus exact closure for unsupported dependencies. We discuss the held-out changed-fraction heuristic separately below.

Table 3. Canonical evidence summary. Work factors are not wall-clock speedups.

	Public RGB/SfM	Trained 3DGS
Cases	60	5
Certified local	44	4
FULL fallback	16	1
Observed violations	0	0
Selected work / FULL	0.36953	0.03495 [†]
Reported reduction	$2.706\times$	$28.614\times$ [†]
Median inspection / FULL	≈ 1.0	0.999997

[†]Trained results use native temporal/output work, not the public calibrated model. Ratios and reduction factors are medians.

7.3 Trained-3DGS representation check

For the secondary check, we use a pinned public trained 3DGS PLY and preserve its spherical-harmonic coefficients, opacity, anisotropic scale, and rotation. Across the five frozen revisions, CBRC chose four certified-local repairs and one full fallback, with no observed violation. Under the native temporal/output work definition used for this experiment, the median selected ratio was 0.03495 of FULL, or about $28.614\times$ lower native work.

The single scalar hides the most interesting part of this experiment, which is why we also report the domain decomposition in Table 4. Median Gaussian update and GPU publication ratios are about 0.0182, and temporal invalidation is about 0.03495. Gaussian inspection, however, remains

Table 2. Frozen public method comparison. The work column is the baseline suite’s native work-ratio metric and is distinct from the calibrated heterogeneous headline ratio of 0.36953. *Uncertified* means the selected cone failed the conservative certificate, not that an independent oracle violation was observed. All four tested radius policies produced the same aggregate values on this matrix.

Method	Cert. pass	Uncertified	FULL rate	Median native work/FULL	Decision rule
FULL	100.0%	0.0%	100.0%	1.00000	full reference
Exact only	26.7%	73.3%	0.0%	0.00000	exact closure only
Radius 0-3	26.7%	73.3%	0.0%	0.00000	fixed graph radius
Changed fraction	40.0%	60.0%	0.0%	0.00000	threshold policy
Empirical order + certifier	100.0%	0.0%	0.0%	0.04736	empirical ordering, analytic recertification
CBRC	100.0%	0.0%	26.7%	0.04736	certificate + work crossover + fallback

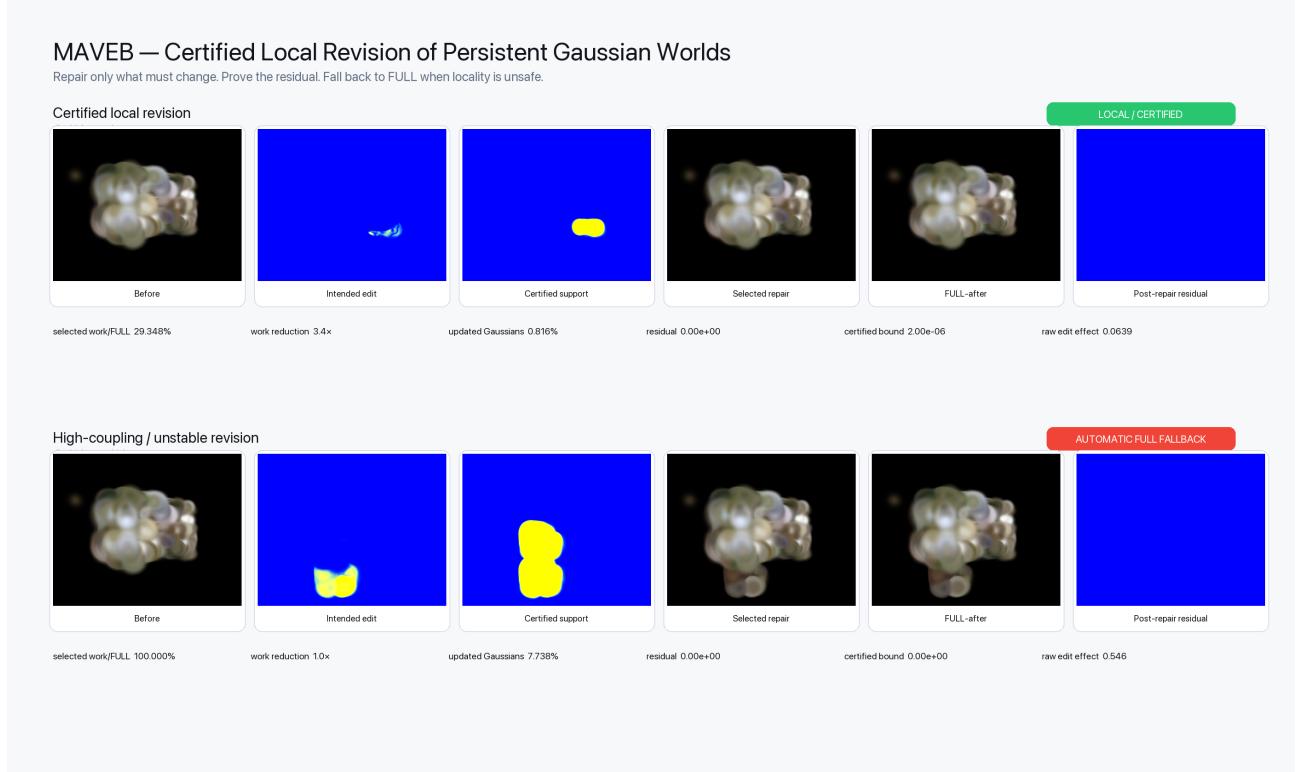


Fig. 3. Representative frozen revision outcomes. Top: a certified local case has zero measured support-replay residual against the independent FULL-after reference at the reported precision. Bottom: a higher-coupling revision expands to an automatic FULL fallback rather than forcing a local decision. The intended-edit and certified-support panels make the source effect and selected scope explicit; the residual panel is computed against the independent full-after oracle.

0.999997 of FULL. In other words, the update itself can be sharply local while finding the candidates is still almost global. We regard that as the main systems limitation of the frozen end-to-end implementation.

Table 4. Median native-domain ratios in the trained-3DGS validation.

Native domain	Ratio / Full
Gaussian inspection	0.999997
Gaussian update	0.018154
GPU publication bytes	0.018154
Temporal pixels invalidated	0.034948

Inspection remains nearly global even when the selected update, publication, and temporal domains are local. The table exposes rather than hides the discovery bottleneck.

7.4 Sparse candidate discovery

We ran a separate exact sparse-discovery sweep to check whether near-full inspection is actually inherent to candidate selection. It is not, at least in this isolated study. Candidate selection remains exactly unchanged while the median inspected fraction falls to 0.01 and the minimum to 0.001, up to the tested scale of one million Gaussians (Table 5). We keep this result separate from the frozen end-to-end ledger because the index was evaluated on its own; crediting those savings to the main pipeline would mix two different experiments.

Table 5. Separate sparse-discovery study summary.

Statistic	Value
Exact candidate-selection agreement	true
Median inspected fraction	0.010
Minimum inspected fraction	0.001
Largest tested scale	1,000,000 Gaussians

The exact sparse-discovery sweep preserves candidate selection while substantially reducing inspected support. These savings are reported separately and are not credited to the frozen end-to-end headline work number.

7.5 Empirical scheduling and ablations

The held-out changed-fraction scheduler did not produce an observed unsafe false-local decision on the frozen matrix. That sounds encouraging, but the result is weak evidence for empirical safety because the candidate full-reference residual is zero across these cases. The matrix therefore does not strongly distinguish empirical scheduling from certification. For this reason, CBRC keeps empirical edges outside the certificate by design.

Table 6 is less tidy than a synthetic story would be. Five structural or analytic ablations are neutral on these 60 real cases. A global-norm tail remains certificate-safe but sends every case to FULL. Removing fallback fails the other way: it never chooses FULL, and only 26.7% of cases pass the certificate. In the separate mechanism-isolation suite, all seven deterministic stress cases move in the expected safety/work direction. Those cases show falsifiability, not real-world effect size.

At the reported precision, every selected public support-replay residual is numerically zero, even though 56 of the 60 source edits were above tolerance before repair. This lets the real matrix test selection and fallback behavior, but it does not tell us how tight the certificate is when a non-zero residual is still acceptable. The threshold-crossing mechanism-isolation cases are useful here only in a limited sense: they show that the analytic machinery can change a decision when it is stressed. We do not use them as a stand-in for real non-zero-residual evidence.

8 DISCUSSION

8.1 What the evidence establishes

Within the frozen revision domain and under the stated bound assumptions, three conclusions are supported by the experiment. The emitted certificate survived independent full-reference falsification of the unrepaired exterior in every evaluated case. The planner also did not force a local answer: more than a quarter of the public revisions crossed to FULL. Finally, when a local result was selected, the frozen calibrated model usually assigned much less work than a full rebuild. This is stronger evidence than showing that a changed-region heuristic works on average, but it is still far short of a proof of universal production safety.

For us, the central distinction is between *certification* and *prediction*. An empirical scheduler can guess every available case correctly and still have no defensible answer for the next unseen case. To become part of the safety argument, that prediction would need a conservative bound with a valid remainder. CBRC therefore lets empirical magnitudes influence scheduling, but does not use them as soft certificate bounds; unsupported empirical-only dependencies are instead closed exactly. The converse can also happen: an ANALYTIC edge may be so conservative that it pushes the system to FULL more often than we would like. That is an efficiency cost of the contract, not a correctness failure.

8.2 Limitations

Bounds are only as sound as their assumptions. The certificate is only as trustworthy as its edge assumptions. Exact dependencies needed to reproduce repaired state must appear in HARD closure, and each ANALYTIC edge must remain a valid finite-change upper bound over the declared revision domain. Representation, shader, temporal-rule, or ownership changes can invalidate earlier calibration, so bound provenance must be versioned and regression-tested.

Gaussian discovery remains global in the frozen path. The clearest implementation limitation is still Gaussian inspection. In the frozen path it is almost a full scan. The sparse-discovery experiment shows that exact selection can be recovered with far fewer inspections, but that index has not yet been integrated into the complete measured pipeline. Until it is, the low update ratios should not be read as end-to-end complexity reduction across every phase of the system.

No wall-clock speedup claim. The public 2.706 $\times$ figure is a reduction in calibrated four-domain work. The trained 28.614 $\times$ figure is a reduction in native temporal/output work. Neither number is a paired local-versus-full runtime speedup. The missing experiment is straightforward in principle: after sparse discovery is integrated, run matched end-to-end timings under identical conditions and report those timings directly.

Greedy search is not globally optimal. Equation 15 gives us a certified feasible cone, not a proof that the cone is the

Table 6. Frozen real-matrix ablations and their deterministic mechanism-isolation interpretation. Native work ratios use the baseline-suite metric. Neutral means the real matrix matches CBRC on the reported aggregate fields; it is not claimed as evidence that the removed mechanism is unnecessary.

Ablation	Pass	FULL rate	Median native work/FULL	Reading
Remove predecessor closure	100.0%	26.7%	0.04736	neutral on real matrix; synthetic case exposes omitted required exact work
Gaussian analytic $\rightarrow$ HARD	100.0%	26.7%	0.04736	neutral on real matrix; synthetic case increases required work or forces FULL
Temporal analytic $\rightarrow$ HARD	100.0%	26.7%	0.04736	neutral on real matrix; synthetic case increases required work or forces FULL
Remove QoI specialization	100.0%	26.7%	0.04736	neutral on real matrix; synthetic case requires extra irrelevant work
Collapse HARD/ANALYTIC separation	100.0%	26.7%	0.04736	neutral on real matrix; synthetic case becomes more conservative
Global-norm exterior tail	100.0%	100.0%	1.00000	safe but destroys locality on all 60 real cases
Remove certified fallback	26.7%	0.0%	0.00000	44/60 real cases fail certification; synthetic case leaves an unsafe candidate

cheapest safe one. Finding the global minimum-work solution is a combinatorial optimization problem and is outside the v1 claim.

Analytic cycles fail closed. The reference formulation allows the classical resolvent when its assumptions are satisfied. The native v1 implementation is narrower and certifies finite analytic DAG exteriors. Unsupported cycles therefore cause the cone to expand or fall back to FULL. General cyclic certification remains future work.

Representation and dataset scope. The primary public campaign is built from normalized RGB/SfM reconstructions seeded into Gaussian fields, not from trained photorealistic 3DGS, and it is not a metric-scale reconstruction benchmark. Its frozen evidence exercises Gaussian inspection/update, GPU publication, and temporal output invalidation; the broader observation/TSDf/mesh/texture/material ledger and captured-world graph are implemented and regression-tested but are not measured by the 60-case headline campaign. The trained-3DGS experiment helps, but it is only a five-revision representation check over the same output-side scope. A broader full-stack evaluation over trained scenes and additional derived-state domains is needed for stronger evidence of generality.

Ablation identifiability. Not every mechanism separates on the frozen real matrix. Rather than forcing a cleaner story, we keep the real-workload results distinct from the synthetic mechanism-isolation results. The paper therefore does not claim that every mechanism produces a measurable benefit in every scene.

8.3 Implications

The broader point is not that persistent captured worlds should always update locally. Locality can instead become a runtime decision. If dependencies can be separated into exact relations and conservatively bounded ones, a local update can carry a falsifiable output contract rather than relying on a

heuristic alone. The same principle can extend beyond Gaussian rendering to caches, temporal histories, spatial acceleration structures, compressed representations, and publication buffers when their dependency semantics are explicit.

Failure is also more useful when it is kept intact. A certificate violation is not folded into an average quality score; we retain it as a regression target together with the revision manifest, graph version, chosen cone, emitted bound, full-reference output, and the first violated invariant. That may be less convenient than a single aggregate metric, but it matches the correctness burden introduced by selective recomputation.

9 REPRODUCIBILITY AND ARTIFACT SCOPE

The canonical evidence freeze records workflow run IDs, artifact IDs, Git SHAs, archive hashes, campaign-row hashes, calibration hashes, and the hash of the pinned trained PLY. The public source archive is fixed by byte size and SHA-256. The trained PLY is fetched from an explicit repository revision and path, with its byte size and SHA-256 checked, and MAVEB does not redistribute it. The repository also contains the dense reference implementation, native planner and oracle, public and trained campaign drivers, baseline and ablation suites, the sparse-discovery sweep, deterministic paper-artifact generators, and regression tests.

Scene figures retain the relevant upstream terms. Original software is licensed under Apache-2.0, original research communication is scoped under CC BY 4.0, and third-party rights are recorded separately. For double-blind review, the identifiable repository and author metadata should be replaced with an anonymized frozen supplement that follows the target venue’s policy.

10 CONCLUSION

We introduced Criticality-Bounded Revision Cones as a fail-closed rule for selective recomputation in persistent captured worlds. Criticality is the worst normalized certificate-to-tolerance ratio over the protected outputs. CBRC separates exact dependencies from bounded finite-change influence,

ties the unrepaired exterior to explicit tolerances, compares regional work with a full baseline, and independently falsifies the residual outside the selected repair support against a full-after oracle. The frozen public and trained-representation experiments show that local repair and full fallback can co-exist under one auditable contract. They also expose the remaining gaps: discovery is still close to global in the frozen path, and there is no paired wall-clock speedup result yet. Repair locally when the system can justify what it leaves alone; rebuild when it cannot.

ACKNOWLEDGMENTS

Generative AI use disclosure. OpenAI ChatGPT was used as an assistive tool during portions of this project for literature-search support, software and scripting development, debugging, experimental and benchmark planning, visualization/manuscript-artifact scripting, and drafting, restructuring, and language editing of the manuscript and supplement. The reported measurements and frozen evidence were produced by the project software and experimental artifacts rather than generated by the language model. The author retains full responsibility for the code, citations, analysis, claims, conclusions, and final text; no generative AI system is listed as an author.

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